

Hotel Bar Inventory Forecasting & Par Level Recommendation

Arun Ananthula | Krystal Ball assignment - managerial summary | data 2023-01-01 to 2024-01-01 (366 days), 6 bars, 16 brands, 96 modelled bar-brand series | back-test 2023-10-20 to 2024-01-01 (74 days), L = 2 days, 95% service level

Key findings

- **Demand is intermittent:** 84% of bar-brand-days have no consumption (~1 event per 6 days), with no material weekday pattern (busiest day 1.04x average). Balances satisfy the conservation equation on 100.0% of rows (after repairing cells the source PDF had rounded).
- **Forecasting:** Linear regression has the lowest lead-time-demand RMSE (205 ml, -11% vs the 7-day moving average). Daily WAPE is above 100% for every model because of the zeros, so weekly-total WAPE and RMSE are used.
- **Par levels:** the textbook Z (1.645) served only 64% of demand events; a Z calibrated on training data (3.59) serves 94.5% against the 95% target.
- **Recommendation:** dynamic par with calibrated Z and a one-bottle minimum order: 19 stockout days vs 41 recorded, with -60% average stock. At equal average stock, dynamic vs static par: 62 vs 59 stockout days (about the same).

1. Core business problem & operational impact

Bars lose money in two opposite ways. **Stockouts:** a guest cannot order a popular brand or signature drink, revenue is lost and service is interrupted. **Overstock:** excess bottles tie up working capital and scarce backroom space and raise shrinkage/spoilage risk. In the history analysed, **5,261 of 34,769 bar-brand-days (15.1%)** ended with zero stock (worst bar: Thomas's Bar, 19.1%), and on **264 of 6,575 recorded consumption events (4%)** the stock ran out while serving, so recorded consumption understates true demand (indicative unserved volume ~245 litres). Meanwhile the median class-A series holds 29.2 days of cover versus 50.6 for class C. Slow movers are over-covered while fast movers run out. There is no material weekday pattern (busiest day only 1.04x an average day). Demand is **intermittent**: 84% of series-days have no consumption (one demand event every ~6 days per bar-brand), so the risk lies in the size of the next demand event rather than in the day of the week.

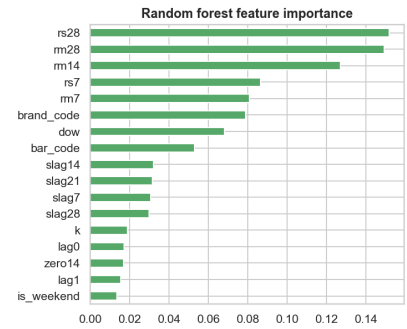
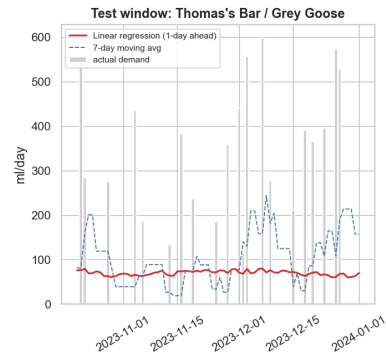
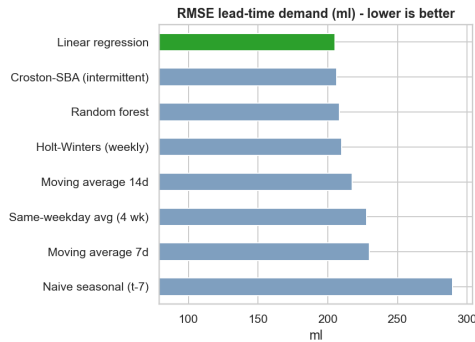
2. Assumptions made

- Supplier lead time is constant at L = 2 days (sensitivity 1-3 days tested) and the supplier always has stock; unmet demand is **lost, not back-ordered**.
- Recorded *Consumed* is demand; when stock runs out demand is censored (true demand was higher). Quantities are in ml (a standard bottle is assumed to be 750 ml); par levels and order quantities are in ml and not rounded to bottles, because typical daily use is a small fraction of a bottle.
- Consumption is aggregated to calendar days per bar-brand with explicit zero-demand days; only pairs that really occur are modelled (96 of 96; the rest have too little history). The balance equation Closing = Opening + Purchase - Consumed holds on 100.00% of rows; violations are flagged and kept, duplicates dropped.
- Replenishment is order-up-to par with a daily review, starting stock = par, no case-size constraint (a one-bottle minimum order is tested as a variant). The safety factor Z is **calibrated on training data** for the 'serve 95% of demand events' target instead of assuming normal errors, because 84% zero-demand days make errors highly skewed; the forecast-error sigma uses a trailing 56-day window.

3. Model selection & trade-offs

Compared on a strict temporal split (first 80% of days train, last 20% test) and on identical (day, series) cells: naive seasonal, moving averages, same-weekday average, Croston-SBA, Holt-Winters (weekly), linear regression and a global random forest with lag / rolling / weekday features. Because 84% of series-days are zero, an always-zero forecast scores exactly 100% daily WAPE while being useless, so daily WAPE (best model: 166%) is inflated for every model. We therefore also report WAPE on weekly totals (78% for Linear regression vs 63% for the 7-day moving average) and **rank models on the RMSE of lead-time demand**, the error that sets the safety stock. **Linear regression** was chosen (205 ml vs 230 ml for the 7-day moving average, -11%); it is fast and interpretable with lag and weekday features; trade-off is that it cannot model interactions such as weekend x trend. Selection was made on the same test window, so a production version should add rolling-origin validation.

Model	WAPE next day	WAPE weekly totals	RMSE lead-time demand (ml)	Bias next day
Naive seasonal (t-7)	175%	100%	289	+0.4%
Moving average 7d	166%	63%	230	+0.5%
Moving average 14d	167%	74%	218	+1.3%
Same-weekday avg (4 wk)	167%	84%	228	+0.7%
Croston-SBA (intermittent)	168%	80%	207	+1.2%
Holt-Winters (weekly)	167%	76%	210	+0.9%
Linear regression	166%	78%	205	-1.3%
Random forest	173%	82%	208	+8.9%
Zero forecast (reference)	100%	100%	234	-100.0%



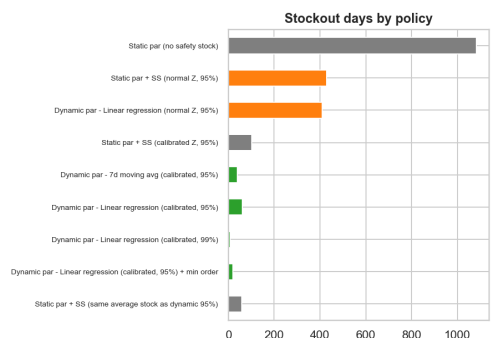
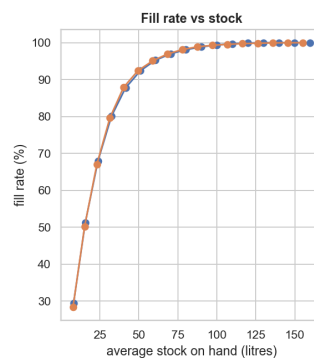
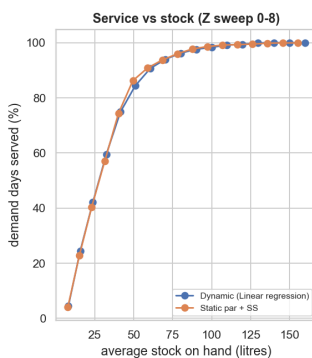
4. Performance & future improvements

Forecast accuracy (test window): lead-time-demand RMSE 205 ml for Linear regression vs 230 ml for the 7-day moving-average baseline (-11%); weekly-total WAPE 78% vs 63%; next-day WAPE 166% vs 166% (an always-zero forecast scores 100%).

Simulation (74 days, L = 2, target: serve 95% of demand events):

- **Textbook vs calibrated Z.** Z = 1.645 served only 63.7% of demand events (410 stockout days) because zero-heavy demand is far from normal; Z calibrated on the training window (3.59 for 95%, 5.72 for 99%) served 94.5% (99.2% at 99%).
- **Versus recorded history** (41 demand days where stock ran out in the same window - indicative, as recorded demand is capped there - with 270 L on hand): calibrated dynamic par had 62 stockout days with 72 L (-73% stock). **With a one-bottle minimum order (recommended):** 19 stockout days, fill rate 99.3%, 108 L (-60% stock) and 581 orders vs 315 recorded deliveries (1,966 for the pure order-up-to rule).
- **Dynamic vs static par at the same average stock:** 62 vs 59 stockout days, lost volume +10%. At equal average stock, dynamic and static par perform about the same (62 vs 59 stockout days): this demand is stationary and intermittent with no weekday pattern, so a forecast-driven par adds little over a well-calibrated static par. Its value is automatic re-calibration when demand shifts (new brands, events, trends); the big gains here come from calibrated Z and right-sized stock. A naive static par without safety stock had 1,084 stockout days.

Policy	Stockout days	Demand days served	Fill rate	Lost volume (L)	Avg stock (L)	Turnover (x/yr)
Static par + SS (calibrated Z, 95%)	102	91.0%	95.3%	19	60	32.7
Static par at same avg stock as dynamic 95%	59	94.8%	97.4%	10	72	27.1
Dynamic par, 7-day MA (calibrated, 95%)	40	96.5%	98.0%	8	86	22.7
Dynamic par, Linear regression (textbook Z 1.645)	410	63.7%	82.7%	68	35	55.6
Dynamic par, Linear regression (calibrated, 95%)	62	94.5%	97.2%	11	72	27.1
Dynamic par, Linear regression (calibrated, 99%)	9	99.2%	99.8%	1	114	17.1
Dynamic 95% + 750 ml min. order (recommended)	19	98.3%	99.3%	3	108	18.0
Historical (actual)	41	96.4%	n/a	n/a	270	7.2



Next steps: promotion / event / holiday calendar features; live POS event feed instead of end-of-day logs; censored-demand correction after stockouts; probabilistic (quantile) forecasts and supplier lead-time distributions; joint vendor delivery consolidation with a weekly review period and case-size / minimum-order constraints; rolling-origin cross-validation for model selection.

5. Real-world deployment & production considerations

- **Pipeline:** a nightly cron job ingests the day's balance logs, runs validation (conservation equation, duplicates, unit sanity), refreshes features, forecasts the next L days, recomputes par per bar-brand and writes a manager dashboard / CSV: on hand, par, forecast, *recommended order quantity* and stockout-risk alerts. Weekly refit of the ML model; a moving-average fallback if the job or model fails.
- **What breaks at scale:** supplier lead-time variance (a late delivery erodes safety stock), holiday and event shifts, pouring waste / breakage / comps that make balances drift from sales, new brands with no history, chronic stockouts that hide true demand, and per-series fitting cost as bars and SKUs grow (the global model scales better than per-series Holt-Winters).
- **Monitoring:** rolling WAPE and bias per bar and class vs the baseline, data-drift / level-shift detection on demand trends, achieved service level vs target, forecast MSE vs real orders, share of rows failing the balance check, and manager overrides as a feedback signal.